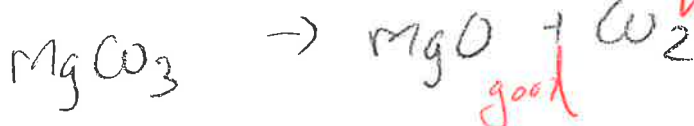
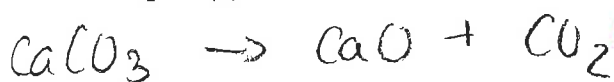


Limestone is available with the composition 90 wt% calcium carbonate (CaCO_3), 8 wt% magnesium carbonate (MgCO_3), and 2 wt% Insoluble Inerts (I). The limestone is heated to 1000°C in a continuous rotary lime kiln to drive off carbon dioxide (CO_2) and make lime. Lime will include the CaO and MgO produced plus all the impurities present in the limestone which remain after all the CO_2 has been driven off. Assume that both the calcium carbonate and the magnesium carbonate react completely to produce carbon dioxide.

Data: $M_{\text{CaCO}_3} = 100 \text{ kg/kmol}$, $M_{\text{MgCO}_3} = 84.3 \text{ kg/kmol}$, and $M_{\text{CO}_2} = 44 \text{ kg/kmol}$.

- (a) What is the fractional atom economy for the process if lime is the desired product? (Hint: You can choose any convenient basis for your calculation, such as, 100 lb_m of limestone.)
- (b) If limestone costs \$87/ton and lime can be sold for \$125/ton, what is the profit (or loss) rate (\$/day) for a lime kiln producing 250 ton/day of lime? Be sure to indicate whether it is profit or loss. (Hint: How much limestone is required to make 250 ton/day of lime?) 1 ton = 2000 lb_m
- (c) If for this process, it costs \$75/ton of carbon dioxide for the proper disposal of CO_2 waste, how does this change the result in part (b)?

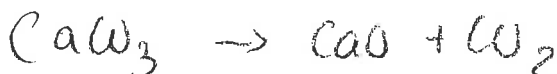


a) Let the basis be 100 kg limestone

$$\Rightarrow \text{CaCO}_3 = 90 \text{ kg}$$

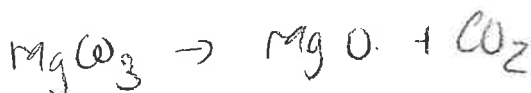
$$\text{MgCO}_3 = 8 \text{ kg}$$

$$\text{inert} = 2 \text{ kg}$$



$$\text{mass fraction: } \frac{56}{100} = 0.56$$

$$\Rightarrow m_{\text{CaO}} = 90 \times 0.56 = 50.4 \text{ kg}$$



$$\Rightarrow \text{mass fraction} = \frac{40}{84.3}$$

$$\Rightarrow m_{\text{MgO}} = 8 \times \frac{40}{84.3} = 3.8 \text{ kg MgO}$$

$\Rightarrow$ total lime produced:

$$m_{\text{lime}} = 50.4 + 3.8 + 2 = 56.22 \text{ kg}$$

$$\Rightarrow \text{FAE} = \frac{56.22}{100} = 0.562$$

(Fraction atom economy) yes! = 56.2%

b) From part a:

$$100 \text{ kg limestone} \rightarrow 56.22 \text{ kg lime}$$

$\Rightarrow$ limestone required =

$$250 \times \frac{100}{56.22} = 444.8 \text{ ton/day}$$

Cost:

$$444.8 \times 87 = 38697.6 \text{ \$/day}$$

Revenue from lime

$$250 \times 125 = 31250 \text{ \$/day}$$

$$\text{Profit} = 31250 - 38697.6$$

$$= -7447.6 \text{ \$/day}$$

$\Rightarrow$ The process operates at loss of \$7448/day

$$c) \text{ CO}_2 \text{ from } \text{CaCO}_3 : 90 \times \frac{44}{100} = 39.6 \text{ kg CO}_2$$

$$\text{CO}_2 \text{ from mag CO}_3 : 8 \times \frac{44}{84.3} = 4.18 \text{ kg CO}_2$$

$$\Rightarrow \text{total CO}_2 \text{ per } 100 \text{ kg limestone} = 39.6 + 4.18 = 43.78 \text{ kg CO}_2$$

$$\text{Daily CO}_2 \text{ produced} = 444.8 \times 0.4378 = 194.8 \text{ ton CO}_2/\text{day}$$

$$\text{CO}_2 \text{ disposal cost} = 194.8 \times 75 = 14610 \$/\text{day}$$

$$\text{New profit} = -7468 - 14610 = -22078 \$/\text{day}$$

good work ✓